BOUNDEDNESS AND CONVERGENCE OF THE SPECTRAL SEQUENCE OF A FINITELY FILTERED COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all four assertions; the sentence 'Part (2) is clear from Equation ...' refers to a displayed identity established earlier in the chapter rather than declining a step, and it is preserved verbatim together with the rest of the proof.

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Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Filtered complexes and the filtration on cohomology. Same chapter, Definition definition-filtered and the construction of the filtration on H^n(K). Supplies the notation of the statement and the identity from which finiteness of the filtration on cohomology is read off.
- The weak convergence criterion. Same chapter, Lemma lemma-filtered-complex-ss-converges together with the displayed descriptions of the top and bottom pieces. Reduces convergence to two displayed equalities, which the proof verifies using the finiteness of the filtration.
- Boundedness implies regularity and coregularity. Same chapter, Lemma lemma-relate-boundedness. Supplies that a bounded spectral sequence is regular, which gives convergence, and coregular, which lets the author replace a high page by the limiting page.
- Weak Serre subcategories are closed under extensions. Same chapter, Lemma lemma-characterize-weak-serre-subcategory. Supplies that the middle term of a short exact sequence with outer terms in a weak Serre subcategory again lies in it, applied to the finite filtration of H^n.

Date: stacks-project:homology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 0121: sources/stacks-homology/pinned-original-source.tex, source lines 4884–4884. Navigation pointer invoked at proof line 6582; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 012R: sources/stacks-homology/pinned-original-source.tex, source lines 6392–6392. Navigation pointer invoked at proof line 6596; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 012V: sources/stacks-homology/pinned-original-source.tex, source lines 6559–6559. Navigation pointer invoked at proof line 6597; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 012S: sources/stacks-homology/pinned-original-source.tex, source lines 6451–6451. Navigation pointer invoked at proof line 6565; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 012T: sources/stacks-homology/pinned-original-source.tex, source lines 6461–6461. Navigation pointer invoked at proof line 6565; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BDV: sources/stacks-homology/pinned-original-source.tex, source lines 6509–6509. Navigation pointer invoked at proof line 6609; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0754: sources/stacks-homology/pinned-original-source.tex, source lines 2366–2366. Navigation pointer invoked at proof line 6619; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $(K^\bullet, F)$ be a filtered complex of $\mathcal{A}$. Assume that the filtration on each K^n is finite (see Definition 0121). Then*

1. *the spectral sequence associated to $(K^\bullet, F)$ is bounded,*
2. *the filtration on each $H^n(K^\bullet)$ is finite,*
3. *the spectral sequence associated to $(K^\bullet, F)$ converges to $H^*(K^\bullet)$,*
4. *if $\mathcal{C} \subset \mathcal{A}$ is a weak Serre subcategory and for some r we have $E_r^{p,q} \in \mathcal{C}$ for all $p, q \in \mathbf{Z}$, then $H^n(K^\bullet)$ is in $\mathcal{C}$.*

Proof. Part (1) follows as $E_0^{p,n-p} = \mathrm{gr}^p K^n$. Part (2) is clear from Equation (012R). We will use Lemma 012V to prove that the spectral sequence weakly converges. Fix $p, n \in \mathbf{Z}$. The right hand side of (012S) is equal to $F^pK^n \cap \mathrm{Ker}(d) + F^{p+1}K^n$ because $F^{p+r}K^n = 0$ for $r \gg 0$. Thus (012S) is an equality. The left hand side of (012T) is equal to $F^pK^n \cap \mathrm{Im}(d) + F^{p+1}K^n$ because $F^{p-r+1}K^{n-1} = K^{n-1}$ for $r \gg 0$. Thus (012T) is an equality. Since the filtration on $H^n(K^\bullet)$ is finite by (2) we see that we have abutment. To prove we have convergence we have to show the spectral

sequence is regular which follows as it is bounded (Lemma 0BDV) and we have to show that $H^n(K^\bullet) = \lim_p H^n(K^\bullet)/F^pH^n(K^\bullet)$ which follows from the fact that the filtration on $H^*(K^\bullet)$ is finite proved in part (2).

Proof of (4). Assume that for some $r \geq 0$ we have $E_r^{p,q} \in \mathcal{C}$ for some weak Serre subcategory $\mathcal{C}$ of $\mathcal{A}$. Then $E_{r+1}^{p,q}$ is in $\mathcal{C}$ as well, see Lemma 0754. By boundedness proved above (which implies that the spectral sequence is both regular and coregular, see Lemma 0BDV) we can find an $r' \geq r$ such that $E_\infty^{p,q} = E_{r'}^{p,q}$ for all p, q with $p + q = n$. Thus $H^n(K^\bullet)$ is an object of $\mathcal{A}$ which has a finite filtration whose graded pieces are in $\mathcal{C}$. This implies that $H^n(K^\bullet)$ is in $\mathcal{C}$ by Lemma 0754. □